



Genomic features of a high-risk *mcr-1.1*-positive *Escherichia coli* ST10 isolated from cattle farm environment

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Abstract

The environment plays an important role in the dissemination of clinically relevant antimicrobial-resistant bacteria and genes. In this study, we described genomic features of a plasmid-mediated colistin-resistant *mcr-1*-positive *Escherichia coli* strains (PK-3225) isolated from a dairy farm wastewater sample. After initial isolation and PCR detection of *mcr-1*-positive *E. coli*, whole-genome sequencing was performed using Illumina Hiseq 2500 followed by in silico analysis. Genetic context surrounding the *mcr-1* gene was determined and SNP-based phylogenomic analysis was performed. Furthermore, plasmid analysis and conjugation assays were performed to determine transferability of *mcr-1*. *E. coli* PK-3225 belonged to ST10 and carried a broad resistome that included colistin (*mcr-1*), beta-lactam (*bla*_{TEM-1B}), tetracycline (*tetB*), phenicol (*catA1*), macrolide (*mdfA*), trimethoprim (*dhfrA17*), aminoglycosides (*aadA5*, *aph(3'')-Ib*, *aph(6)-Id*), and sulphonamide (*sul2*) resistance genes. The draft genome of *E. coli* calculated as 4.9 Mbp. Conjugation experiment showed successful transfer of the *mcr-1* gene to *E. coli* recipient strain J53. In silico analysis showed that *mcr-1* was located on IncI2 plasmid of > 59 kb in length, with the *nikB-mcr-1-pap2* gene array, and lack *ISApI1*. The phylogenomic analysis revealed that the PK-3225 was closely related to human ST10 *E. coli* from Brazil and USA. To our knowledge, this is the first draft genome sequence of *mcr-1* carrying *E. coli* isolated from the farm environment in Pakistan. Considering the high burden of colistin resistance in Pakistan, presence of pandemic high-risk *E. coli* clones in the environment requires strict surveillance.

Keywords Farm environment · Wastewater · Antimicrobial resistance · *E. coli* · Comparative genomics

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Introduction

Polymyxins are the small lipopolypeptide of the molecular weight 1200 Da with a polycationic ring attached to the hydrophobic fatty acids (Kola et al. 2016). Polymyxins including colistin are the well-organized drugs and have regained significant interest due to the increasing incidence of multidrug-resistant (MDR) gram-negative bacterial infections (Poirel et al. 2017). The mechanism of colistin is mediated by binding to the lipopolysaccharides of gram-negative bacteria and subsequently disintegrates the membrane. Colistin (Polymyxin E) is known as a last-resort antibiotic to treat MDR gram-negative bacterial infections in clinical use (Paterson et al. 2019). Colistin-resistant Enterobacteriaceae are globally recognized as an important threat for public health (Yao et al. 2016). The discovery of plasmid-mediated colistin resistance gene *mcr-1* in China in 2015 (Liu et al. 2016) and worldwide dissemination of *mcr-1* in bacteria through conjugation has been found a great threat to the public. To date, ten variants of the *mcr-1* gene

(*mcr-1* to *mcr-10*) have been identified in different bacterial species isolated from humans, food, animals, farms, and the environment (Hussein et al. 2021). Colistin resistance gene *mcr-1* has been found on plasmids of different incompatibility types IncX4, IncI2, IncY, and IncHI2 (Li et al. 2017). In Pakistan, *mcr-1* has been identified in *E. coli* isolates from wildlife (Mohsin et al. 2016), human (Mohsin et al. 2017), healthy broiler, and sick broilers (Azam et al. 2020; Rafique et al. 2020). However, to the best of our knowledge, *mcr-1* has not been reported from the farm environment.

Materials and methods

A total of 100 cattle samples were screened along with farm wastewater samples collected in 2019 from 4 corporate commercial dairy farms in Punjab, Pakistan. The loopful inoculum was streaked on MacConkey agar (Oxoid, UK) supplemented with colistin 2 µg/ml for 18–24 hours at 37°C. In this regard, one colistin-resistant *E. coli* isolate (PK-3225 strain) was recovered from a dairy farm wastewater. Antimicrobial susceptibility profile was determined by disk-diffusion and/or microdilution broth methods (EUCAST 2021; CLSI 2021; CLSI 2020). DNA of PK-3225 was extracted using DNeasy blood and tissue kit (Qiagen, Hilden, Germany). Conjugation experiment was performed to determine the transferability of colistin resistance using sodium azide-resistant *E. coli* J53 as recipient strain. The sequencing library was prepared using an Illumina Nextera XT DNA Library with 2 × 150-bp paired-end reads on an Illumina HiSeq 2500 instrument (Illumina, San Diego, CA, USA). De novo assembly was performed using SPAdes v3.9 (Wick et al. 2017) and annotation of the genome was performed using DFAST (<https://dfast.nig.ac.jp/>) (Tanizawa et al. 2019). The sequence was submitted to NCBI/ENA/DBJ (Arnemann 2018). The tools of the Center for Genomic Epidemiology (<http://www.genomicepidemiology.org>) were used for the determination of multilocus sequence type (MLST) (Larsen et al. 2012), acquired antibiotic resistance genes (Zankari et al. 2012), plasmid replicon typing (Carattoli et al. 2014), and serotyping (Joensen et al. 2015) using MLST v2.0, ResFinder v4.1, VirulenceFinder v2.0, and SerotypeFinder v2.0, respectively. The complete sequence of the *mcr*-encoding plasmid was performed using Geneious R9 (Biomatters) and Bandage assembly graph visualizer (<https://github.com/rrwick/Bandage>), using an interactive anchoring of short-reads and contigs alignment combined approach, as previously described (Fuga et al. 2021). Plasmid comparative analysis was performed using the Proksee platform (<https://beta.proksee.ca/>) and BLASTN. The genomic sequences of 103 *E. coli* strains belonging to ST10 were downloaded from the Enterobase *Escherichia/Shigella* database (<https://enterobase.warwick.ac.uk/>) and submitted to antimicrobial resistance gene

prediction by using ABRicate (<https://github.com/tseemann/abricate>) with Resfinder database. All sequences were aligned to the reference sequence using CSI-phylogeny v 1.4 and the phylogenetic tree was constructed based upon the SNP content. The tree was visualized using iTOL v6 (<https://itol.embl.de>) and resistome information based upon ABRicate analysis was added into phylogenetic tree.

Results and discussion

The antimicrobial susceptibility profile showed that the PK-3225 *E. coli* was resistant to ampicillin, cefotaxime, gentamicin, tetracycline, sulfamethoxazole/trimethoprim, tylosin, and colistin (MIC = 4 mg/L), remaining susceptible to moxifloxacin, enrofloxacin, and meropenem. Conjugation experiment showed that *mcr-1* was transferable to sodium azide-resistant *E. coli* J53. De novo assembly result generated a total of 165 contigs, with the total length of 4957826 bp and the GC contents were 50.43%.

The PK-3225 carried antibiotic resistance genes against beta-lactams (*bla*_{TEM-1B}), tetracycline (*tetB*), phenicol (*catA1*), colistin (*mcr-1.1*) macrolides (*mdfA*), trimethoprim (*dfrA17*), aminoglycosides [*aadA5*, *aph(3'')-Ib*, *aph(6)-Id*], and sulphonamide (*sul2*), as well as chromosomal mutations in *gyrA* (S83L, D85L) and *parC* (S80I). Plasmid incompatibility types IncFIB, IncFIC(FII), IncI2, and IncQ1 were predicted in PK-3225. The strain belonged to the O101:H10 serotype and carried multiple virulence genes *cvaC*, *etsC*, *gad*, *hlyF*, *hra*, *iha*, *iroN*, *iss*, *iucC*, *iutA*, *mchB*, *mchC*, *mchF*, *ompT*, *sitA*, *terC*, and *traT*.

The *mcr-1.1* gene was found to be encoding by the IncI2 conjugative plasmid, which has been globally associated with *mcr-1* (Ovejero et al. 2017; Sun et al. 2017; Wang et al. 2017). The complete nucleotide sequence of pPK-3225mcr was 59,630 bp and contains 81 coding sequences (CDS) and a G+C average content of 42.26%. The *nikB-mcr-1-pap2* gene array was found in pPK-3225. Interestingly, the same genetic context of *mcr-1* was also found in colistin-resistant *E. coli* from dairy cows in China (Zheng et al. 2019). On the other hand, *nikB* (relaxase) and *pap2* gene cassettes and been identified in *mcr-1*-positive *E. coli* isolates from outpatients, in Chile (Gutiérrez et al. 2019). More recently, it was discovered in China that the *pap2* gene along with *mcr-1* was necessary to reduce colistin susceptibility (Choi et al. 2020). Comparative plasmid analysis using BLASTN revealed that pPK-3225mcr is highly similar (≥ 99% pairwise identity) to other *mcr-1*-harboring IncI2 plasmids identified in *E. coli* and *Salmonella enterica* isolates from humans, livestock, and chicken meat in China, South Korea, and the Netherlands, respectively (Fig. 1).

The colistin-resistant *E. coli* strain PK-3225 belongs to the pandemic MLST sequence type ST10, clonal complex CC10, serotype O101:H10, phylogroup A, and *fimH30* allele. In this

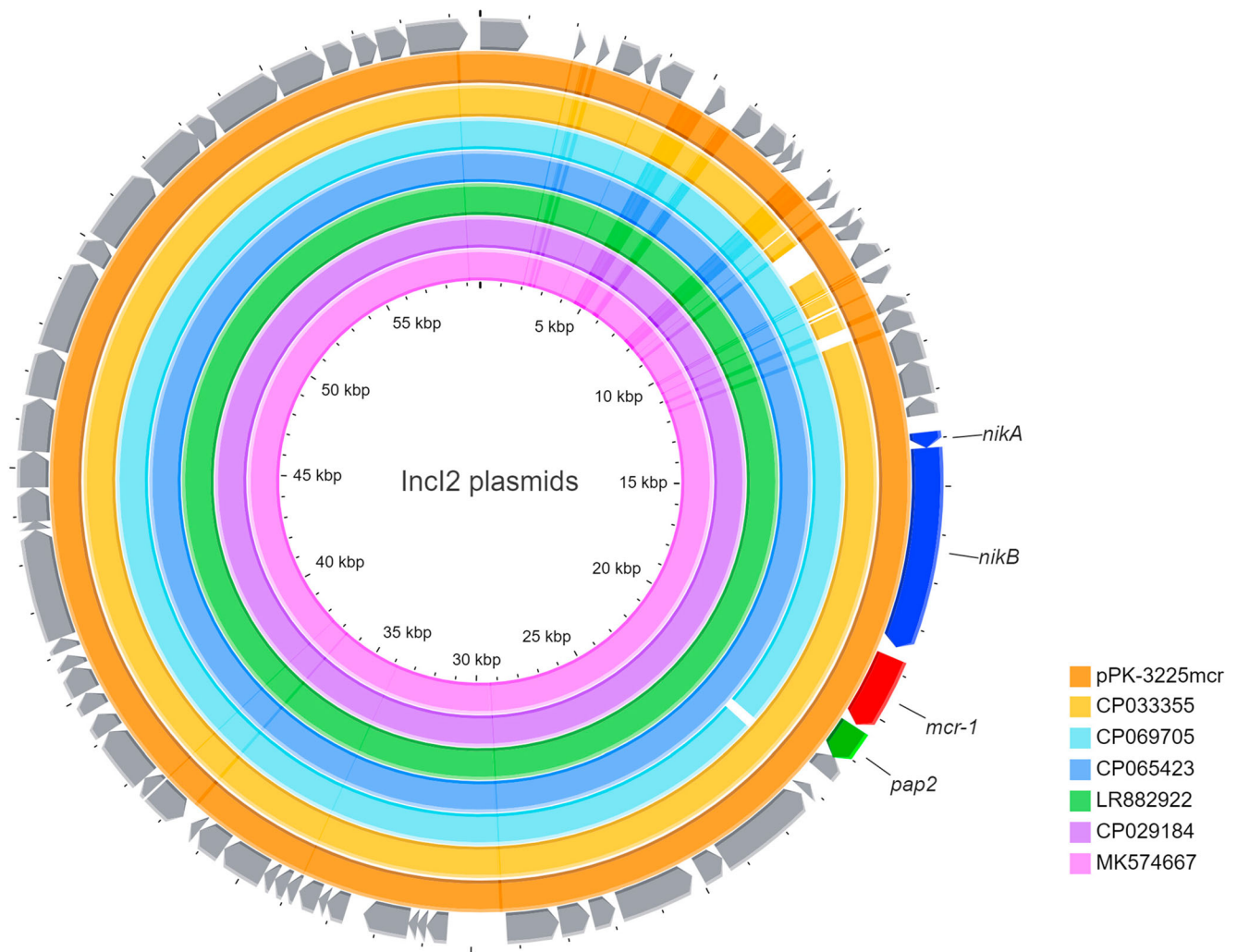


Fig. 1 Circular view and alignment comparison of closely related IncI2 plasmids harboring the *mcr-1* gene. IncI2/*mcr-1* genetic arrays have been identified in *E. coli* strains isolated from farm wastewater in Pakistan (pPK-3225mcr, this study), human hosts in China (MK574667; CP029184) and the Netherlands (LR882922); *Salmonella* Typhimurium

from pig in South Korea (CP065423), and *Salmonella enterica* from chicken meat (CP033355) and *E. coli* from poultry (CP069705) in China. In outer circles, arrows indicate the positions and directions of genes

regard, the adhesin *fimH* allele H30 has been rarely identified in non-ST131 *E. coli* lineages. In fact, *E. coli* ST131-H30 subclade has emerged in recent decades as an epidemic clonal group responsible for a large amount of human extraintestinal infections worldwide, and thus the *fimH* allele H30 could be conferring advantage to human host adaptation (Stoesser et al. 2016). Moreover, virulome analysis revealed the presence of genes related to adherence (*iha*), iron uptake (*iroN*, *sitA*, *iutA*, *iucC*, *hra*), colicins (*cvaC*, *mchBCF*), serum resistance (*iss*, *ompT*, *traT*), hemolysin (*hlyF*), glutamate decarboxylase (*gad*), and tellurium resistance (*terC*).

The phylogenomic analysis of PK-3225 was performed with 103 genomic sequences of *E. coli* strains belonging to the international ST10 (Fig. 2). In this regard, the environmental *E. coli* strain PK-3225 was closely related (SNP difference = 196) to the human *E. coli* strain ESC_FB9722AA, isolated from an urinary tract infection case, in Brazil (Campos et al.

2018) and SNP difference of 1270 from a clinical *E. coli* strain ESC_BA5361AA originated in USA. While *tetB*, *sul2*, *dfxA17*, *bla*_{TEM-1B}, *aadA5*, *aph(3'')-Ib*, and *aph(6)-Id* resistance genes were shared by PK-3225 and ESC_FB9722AA *E. coli* strains, common resistance genes among 103 *E. coli* ST10 genomes analyzed conferred resistance to tetracycline, aminoglycosides, sulphonamide, and trimethoprim (Fig. 2). The presence of the clinically relevant *mcr-1*-positive ST10 *E. coli* in wastewater with multidrug resistance is concerning.

Colistin-resistant *E. coli* strains have been reported from clinical, broiler, and wild migratory bird samples, in Pakistan (Mohsin et al. 2016, 2017; Azam et al. 2017; Lv et al. 2018); however, it has not been reported from farm environments. Therefore, in this study, we report the first draft genome sequence of a multidrug-resistant IncI2/*mcr-1.1*-positive *E. coli* strain belonging to the pandemic ST10, identified in a farm wastewater source in Pakistan. Considering the high burden

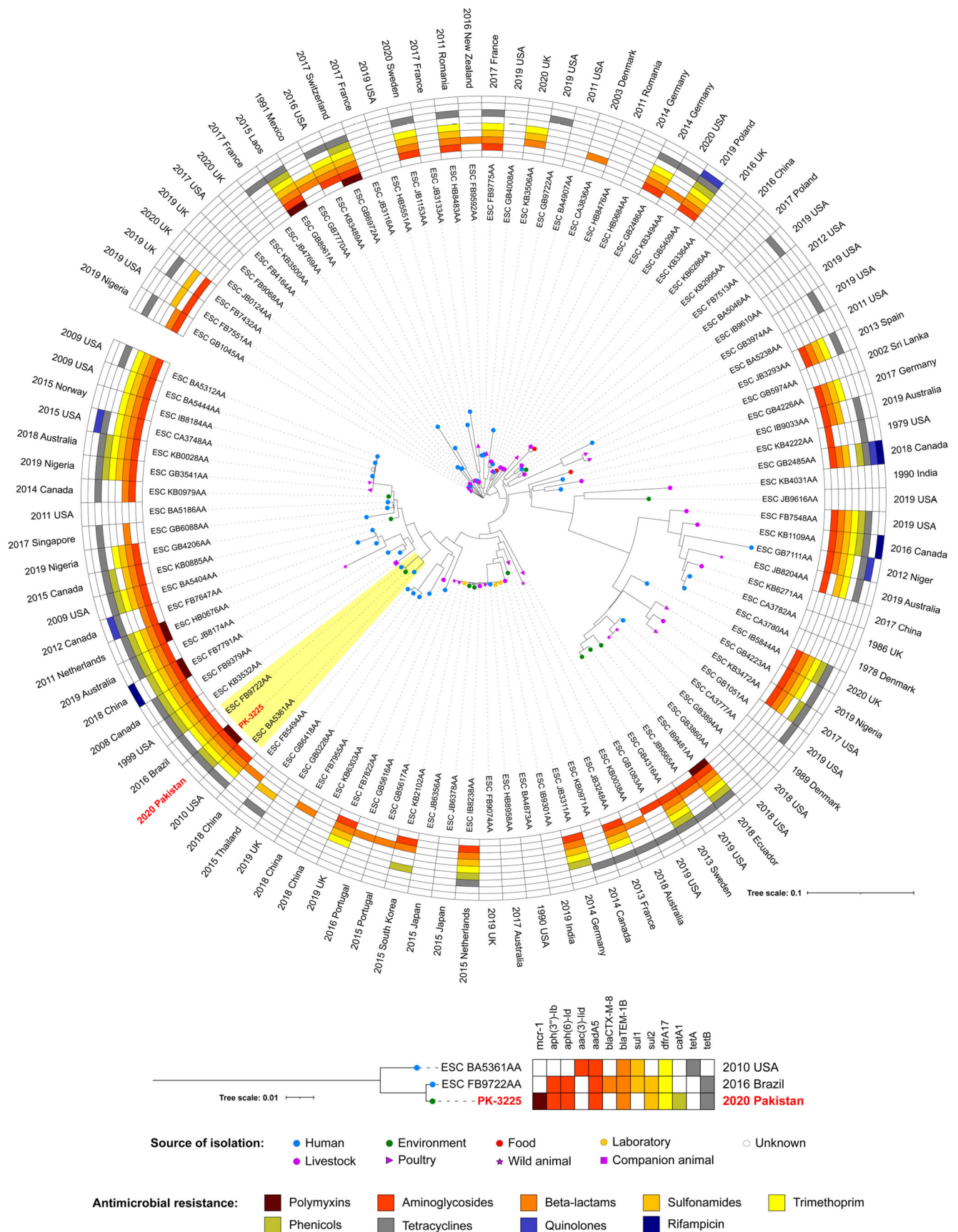


Fig. 2 Phylogenomic comparison of *E. coli* strains belonging to ST10. The maximum likelihood tree was created using CSI-Phylogeny v 1.4 and visualized in interactive tree of life (iTol v6). The PK-3225 was closely related to ESC_FB9722AA Brazil human *E. coli* shown in bold

of colistin resistance in Pakistan, presence of priority high-risk *E. coli* clones in the environment requires strict surveillance.

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Availability of data All data generated or analyzed during this study are included in this published article.

Author contribution AA genomic analysis and wrote the paper, HF and ES analyzed the data, RL reviewed and edited, MH sample collection and PCR, SR reviewed the study, and NL and MM conceptualization and wrote the paper.

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Declarations

Ethical approval and consent to participate Not applicable.

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Competing interests The authors declare no competing interests.

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